A NOTE FROM PRODUCTION

The demo was the easy part.

Every agent demo works. Then it meets a prompt injection it can't see, an auditor asking *“why did it do that?”*, an eval suite that's really vibes, and a cloud bill that ends the experiment. The loop was never the hard part — **everything around the loop is.**

Injections

input rails that are a regex and good intentions

Audits

decisions nobody recorded, answers nobody has

Evals

vibes as the quality gate

Bills

production-grade only if you can afford it

THE QUESTION

What belongs between a working loop and a trustworthy agent?

S

What every agent builder already knows

Frameworks hand you the ReAct loop in fifty lines — and leave guardrails, evaluation, governance, deployment, and the learning curve as exercises for the reader. The tutorials stop exactly where production begins.

Q

So the question this repo answers:

Can one open codebase ship the un-demo-able parts — a guardrail stack with a trained model, an eval pipeline calibrated against humans, policy-gated cloud deployment at hobby cost, and the recipes to rebuild it all — in a form you can verify before you trust it?

THE ANSWER

Yes. It's all here — and checkable.

This repo open-sources the four things agent demos leave out, plus the architecture that holds them together. Six sections follow; every claim carries a file path or a command you can run.

1. **Four-layer architecture** — 12 CI gates; illegal imports fail the build
2. **Guardrail ML** — fine-tuned DeBERTa-v3, frozen eval gates
3. **Goal-judge evals** — grounded theory, κ = 1.0
4. **Free web search** — $0 self-hosted SearXNG, no API key
5. **GCP deployment** — policy-as-code, idles near $0
6. **40+ recipes** — rebuild every piece yourself

108K+ lines · 2,500 tests · 0 live LLM calls in CI · built solo in 8 weeks

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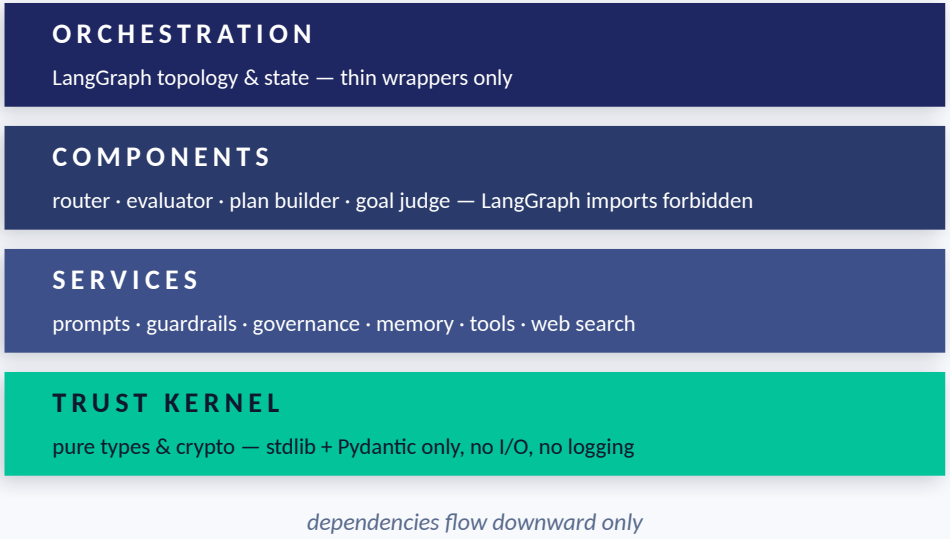
SECTION 1 · ARCHITECTURE

Four layers — crossing them fails the build

S Every framework draws a layering diagram.

C Nothing stops next week's PR from importing across it. Diagrams rot; nobody notices until refactor day.



A **The boundary is a test, not a convention**

12 architecture-gate modules check dependency rules, service isolation, and layer placement on every commit. 2,500 tests overall — a 2:1 test-to-code ratio, with zero live LLM calls in CI.

Verify: *pytest tests/architecture/ -q*

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One task, five phases — not one hopeful loop

S A bare ReAct loop is one pass: if the model misreads the task, the task fails.

C Production needs a gate before the spend and a quality check after — thinking and checking, not just doing.

1 · INGRESS
`guard_input_node`
Reject unsafe input at Step 0 — before any token spend.

›

2 · PLANNING
`route_node`
Pick depth (L0–L2); build a deterministic plan floor.

›

3 · EXECUTION
`call_llm / supervisor`
T0 ReAct spine, or T3 fan-out for independent work.

›

4 · EVALUATION
`evaluate_node`
GoalJudge scores the result; T2 reflexion if it falls short.

›

5 · EXIT
`reasoning_recap`
Synthesize a human-readable 'why' for the UI — or escalate.

All five are real nodes in **`orchestration/react_loop.py`** verify: `grep -n "def .*_node" orchestration/react_loop.py`

Two orthogonal axes, one unbreakable floor

S Most loops use one strategy for every task — over-planning the easy, under-planning the hard.

C And when an LLM plan is malformed, a loop with no backstop simply stalls.

	L0 max 1 step	L1 max 3 steps	L2 max 5 steps
T1 plan artifact	Yes	Yes	Yes
T2 reflexion	Yes	Yes	Yes
T3 fan-out	Never	if ≥2 steps	if ≥2 steps

Depth sets the step budget · tiers add mechanisms — independently.

A **The plan floor never depends on the LLM**

build_plan_artifact lays down an ordered, deterministic plan first. The LLM only enriches it — if the model returns nothing usable, the floor still runs. No brittle stalls.

T3 default: `t3_fanout_enabled = False`

Parallelism that refuses to corrupt

S Fan-out is fast — when tasks are truly independent.

C Run coupled steps in parallel (hotel dates that depend on flights) and you corrupt the result, silently.

`Depth = L0?`	→	DECLINE
`Sequential dependent?`	→	DECLINE · GAIA guard
`No plan generator?`	→	DECLINE
`validate_independence fails?`	→	DECLINE
`All checks pass`	→	**FAN_OUT**

Decline is the default on any unproven path.

MEASURED · T3 STRESS CORPUS

1.0 precision

fp = 0 · 10/10 dependent tasks declined

Small N, designed corpus — a discipline that holds on the cases it was tested on, not a population guarantee.

`docs/plans/t3_stage_b_case_walkthrough.md`

From a brittle loop to a tiered reasoning ladder

S A single ReAct loop applies one strategy to every task and stalls on a bad plan.

C Last week: a four-rung ladder that picks effort by need — every rung shadow-first, default OFF, promoted only on evidence.

T1

Plan with a replan gate

PlanGenerator enriches the deterministic floor; rebuilds when a surprising tool result makes the plan stale.

`plan_source`

T2

Reflexion re-entry

decide_reentry critiques a weak result and loops back — with a D1 attempt ceiling and a D3 prose-thrash backstop.

`reflexion_enabled`

T3

Supervisor fan-out

plan_delegations decomposes independent work; validate_independence declines anything coupled (GAIA guard).

`t3_fanout_enabled`

§5

Escalation predicate

decide_escalation composes the re-entry/continue signals into one decision, measured against a precision oracle.

`escalation oracle`

Shadow-first by default → prod graph stays byte-identical until a rung earns promotion. `services/base_config.py`

Memory: recall, capture, full audit trail

S An agent that forgets every session can't personalize or improve.

C This week: durable recall + typed auto-capture — every read, write, and reject leaves a governance carrier.

RECALL

MEMORY_RECALLED

route_node pulls relevant memories above a relevance floor (dedup'd) into the prompt.

AUTO-CAPTURE

MEMORY_STORED

A debounced LLM extractor proposes TypedMemory facts after the run — shadow-only until calibrated.

GOVERN

MEMORY_SUPPRESSED · _CONSOLIDATED

Per-type budgets consolidate; panel rejects soft-suppress and drop from recall.

M

Durable, swappable, and off by default

Mem0MemoryBackend gives durable storage behind the hexagonal port — swap it without touching the loop. The recalled-memories panel and per-chat soft-suppress ship in the UI.

OFF BY DEFAULT

```
memory_enabled = autocapture_enabled = False
```

SECTION 2 · GUARDRAIL FINE-TUNING

The input rail is a trained model, not a regex

S Everyone ships “guardrails” now.

C Most are a regex and a prayer — or an LLM judge on every request: slow, costly, and untested for over-blocking.

1 · PRECHECK

deterministic: length caps, injection regex, base64 decode, entropy check

zero cost · zero flake

2 · CLASSIFIER

fine-tuned DeBERTa-v3 with PIGuard MOF, exported to INT8 ONNX

reject ≥ 0.80 · accept ≤ 0.20

3 · NARROW JUDGE

small LLM, scoped to injection only — reached only when 1 & 2 defer

the expensive path, made rare

A CI freezes the quality bar: malicious recall ≥ 0.95, false-positive rate < 2% (Llama-Guard-3 ≈ 4%), over-defense measured on a held-out NotInject split — with a contamination guard that makes train/test leakage a hard error. **Training script included: scripts/train_injection_classifier.py.**

SECTION 3 · EVALUATION SCIENCE

A goal judge with human-agreement receipts

S Agents narrate success beautifully.

C “Looks right to me” is not an eval. Judges that trust the agent's own story get gamed by fabricated progress.

κ = 1.0

rubric agreement

two blind annotators; gate ≥0.8 (docs/IAA)

α = 0.88

gold-set pilot

Krippendorff's α on 43 items; 101-row sheet at v0.9

16

failure codes

grounded-theory taxonomy across 5 axes

≤ 5%

red-team flip rate

fabricated-progress attacks must not flip verdicts

A Verdicts are grounded in the tool trace, not the agent's narration — and the judge ships default-off, in shadow mode, until precision ≥ 0.90 and flip ≤ 5% gates clear. **The full 7-stage methodology is in .cursor/skills/llm-eval-grounded-theory/ — 62 tests cover the judge alone.**

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SECTION 4 · FREE WEB SEARCH

Web search that costs $0, needs no API key, and sanitizes what it reads

S Every agent needs web search.

C Search APIs meter every call, and raw snippets are an indirect-injection channel straight into your context window.

$0

Self-hosted SearXNG sidecar — localhost or Cloud Run on port 8888, scale-to-zero, aggregating upstream engines. No vendor, no key, no meter.

- **Hexagonal port —** WebSearchProvider protocol; SearXNG adapter + deterministic stub for CI (services/tools/search/)
- **Injection-sanitized —** every title & snippet runs the retrieval rail before re-entering context; modifications logged
- **Fails terminal, not forever —** dead backend → ok=False; no-progress detector (3 repeats, hard stop 5) ends loop-on-dead-tool
- **Tested failure-first —** 15+ cases: HTTP 500, timeout, network error, empty results — written before the success paths

Verify: *docker compose -f docker-compose.searxng.yml up* · *WEB_SEARCH_PROVIDER=searxng*

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SECTION 5 · GCP DEPLOYMENT

Policy-gated deploys that idle at pocket change

S “Production-grade” usually means a real cloud bill.

C So agent projects stay on laptops — and the deployment lessons never get learned.

$5–30/mo

dev tier, mostly free

Cloud Run scale-to-zero + Neon free Postgres + Cloudflare Pages; the bill is basically Cloudflare Pro

13 + 11

Rego + Gherkin gates

no plaintext secrets, no allUsers IAM, SSE cache bypass — 108 conftest + 38 pytest checks, mutation-tested

4 gates

before any apply

tofu validate → tflint → conftest → pytest; then BDD against the resolved plan

A → D

graduation ladder

free tier → HA production; substrate swaps touch only the composition root, zero app code

A Phased deploys (scripts/deploy_gcp.sh) with two human gates, digest-pinned images, CalVer+SHA deploy IDs — and a black-box → Langfuse relay with a dead-letter queue for observability.

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SECTION 6 · LEARN IT, DON'T JUST CLONE IT

40+ recipes: rebuild every piece yourself

S Most repos document what; few teach how.

C Cloning a black box teaches you nothing — and you can't maintain what you couldn't have built.

13

GCP deployment

adapters → cleanup, with cost notes & rollback per step

9

Guardrails

the "five-door" security build, precheck → classifier → judge

7

Governance

the flight-recorder: black box → Langfuse audit trails

3

Goal judge

grounded-theory failure taxonomy → rubric → gold set

6+

TDD hardening

outcome correctness, schema versioning, negative paths

A Every recipe is dual-audience: exact commands for this repo, generic steps for your stack. Bonus: the reasoning prompts that structured this very deck — SCQA, Pyramid Principle, TDD-for-agents — ship in research/, runnable via StructuredReasoning/cli_pyramid.py.

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HANDS ON

Five things to steal in the next ten minutes

1. **Run the architecture gates**
   pytest tests/architecture/ -q
   watch 12 boundary tests defend the layers

2. **Stand up free web search**
   docker compose -f docker-compose.searxng.yml up
   then set WEB_SEARCH_PROVIDER=searxng

3. **Train the injection classifier**
   python scripts/train_injection_classifier.py
   DeBERTa-v3 + PIGuard MOF → INT8 ONNX

4. **Read one recipe end-to-end**
   docs/recipes/guardrails/00_overview.md
   the five-door security build

5. **Borrow the reasoning prompts**
   research/pyramid_react_system_prompt.md
   MECE issue trees + 8 validation checks, framework-agnostic

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THE ANSWER, RESTATED

If this would've saved you a month, it's worth a star.

Architecture
12 CI gates

Guardrail ML
recall ≥ 0.95 gate

Goal-judge evals
κ = 1.0 · α = 0.88

Web search
$0, no API key

GCP deploy
$5–30/mo, policy-gated

Recipes
40+, dual-audience

Honest fine print: pre-1.0, single maintainer, LICENSE pending — until it lands, treat it as look-don't-ship and watch the repo for the license drop. ~20 live-LLM tests fail visibly in the local cache, excluded from CI on purpose (the recipes explain why). Every number here comes from a static audit of the repository.

github.com/rajnishkhatri/AgentsFramework — star it if it earned one · Rajnish Khatri · rajnish.khatri@gmail.com